Task	Status	Depends On
1. Database Design & Setup	Done	
1.1 Install and configure MySQL on the VM	Done	
1.2 Create initial database instance	Done	
1.3 Build initial ER diagram	Done	
1.4 Revise ER diagram based on feedback	Done	
2. Database Implementation	Done	
2.1 Create tables based on the ERD	Done	
2.2 Insert sample data	Done	
2.3 Normalize tables to at least 3NF	Done	
2.4 Revise schema/ERD	Done	
3. SQL Queries & Access	Done	
3.1 Write SELECT queries, single table	Done	
3.2 Implement JOIN queries, multi-table	Done	
3.3 Data retrieval and modifications	Done	
3.4 Build command-line CRUD app, single table	Done	
3.5 Extend command-line CRUD to many-to-many	Done	
4. Web Integration	Done	
4.1 Set up Flask environment	Done	
4.2 Serve a static webpage	Done	
4.3 Build dynamic page via GET form	Done	
5. Flask CRUD - Single Table	Not Started	Task 4
5.1 Build HTML form templates for Create/Update		
5.2 Add Flask routes for Create, Read, Update, Delete		
5.3 Connect routes to MySQL using parameterized queries		
5.4 Test each CRUD operation through the browser		
6. Flask CRUD - Many-to-Many	Not Started	Task 5
6.1 Build HTML templates to add/remove OS device links		
6.2 Add Flask routes for the join-table operations		
6.3 Reuse/adapt single-table CRUD patterns for the junction table		
6.4 Test adding and removing relationships through the browser		
7. Database Considerations	Not Started	Tasks 5, 6
7.1 Create a dedicated least privilege MySQL user for the app		
7.2 Move credentials out of source; confirm .env is git-ignored		
7.3 Add input sanitization/parameterized queries everywhere		
7.4 Add basic form validation		
8. Repository Cleanup	Not Started	None
8.1 Remove unrelated files (HelloWorld.py, stray flask.py, notes.txt)		
8.2 git ignore venv/ and __pycache__/		
8.3 Organize remaining files into a clear structure		
9. Final Presentation / Demo Prep	Not Started	Tasks 6, 7
9.1 Design a walkthrough of the schema and relationships		
9.2 Prepare clean sample data for the demo		

9.3 Rehearse CRUD flow end-to-end

10. Post-Class Portfolio Polish	Not Started	Task 9

10.1 Write a proper README

10.2 Consider deploying the Flask app somewhere persistent

10.3 Tidy commit history

Effort

Medium (2-3 days)

Large (1 week)

Medium (2-3 days)

Small (1 day)

Small (1 day)

Medium (2-3 days)